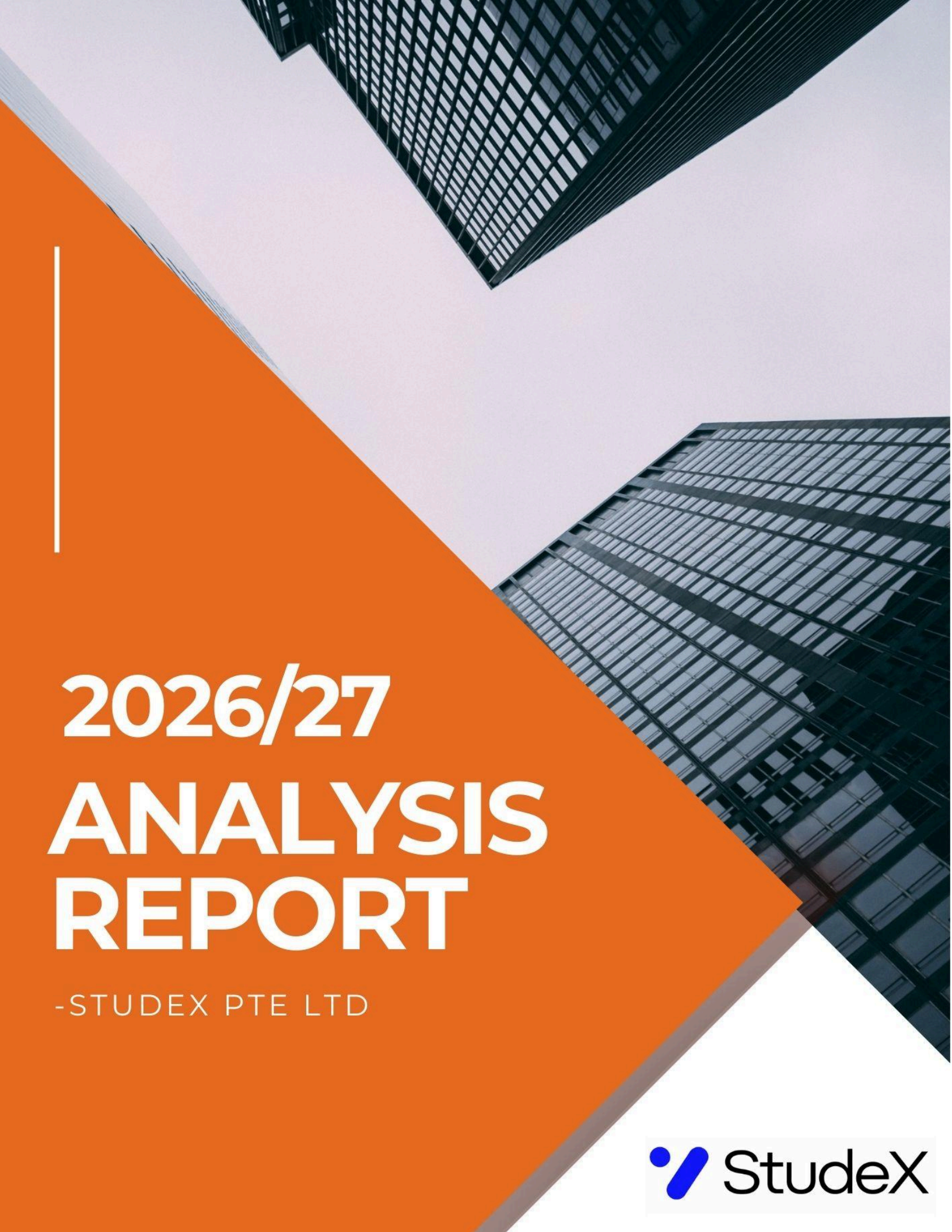

2026/27 ANALYSIS REPORT

-STUDEX PTE LTD

StudeX

Declaration on Non-Plagiarism & AI Tools & Technologies

We, the undersigned, declare that this project is our own collective work and represents our learning from the module. All content, ideas, analysis, and knowledge presented are the result of our own understanding and effort.

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We further declare that AI tools and technologies were used only for grammar correction, spelling improvement, and formatting assistance. AI did not contribute to the development of core ideas, analysis, or substantive content.

We confirm that this submission complies fully with the module guidelines and understand that any violation of these policies may result in a zero mark.

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1. Executive Summary

1.1 Background Information

This report presents Studex Pte. Ltd.'s review of the Project Specification (PS) for the Extracurricular Event Tracking and Participation Management System. The review focuses on assessing scope stability, requirement quality, and readiness for the transition to the Software Requirements Specification (SRS) stage.

Through structured “Breakdown, Clarify, Interpret & Categorise” (BCIC) analysis, we identified several incongruities in the PS, which were then clarified, refined, and decomposed to ensure that the original intent was preserved. An Internal Requirements Checklist (IRC) was applied to ensure consistency, completeness, and testability across the analyzed requirements. Complete Analysis of the requirements quality has been evaluated and presented in this report.

Based on the completed analysis, the PS is now considered fit for progression to the SRS stage, subject to the incorporation of the clarifications and refinements identified in this report. With scope boundaries enforced and change governed through vendor controls, the project is considered institutionally aligned and technically feasible for subsequent specification and design activities.

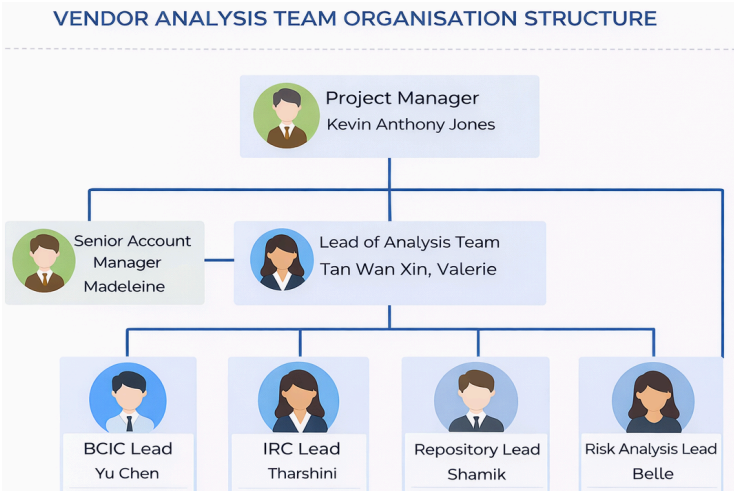
1.2 Project Background

Apex Institute of Technology approached Studex Pte. Ltd. to develop the Extracurricular Event Tracking and Participation Management System to address challenges in managing and tracking student participation in extracurricular activities. The client sought a digital solution that would improve transparency, efficiency, and accountability in tracking student attendance, allocating participation hours, and managing event registrations. Studex was chosen for this project due to our expertise in providing reliable digital solutions for educational institutions.

1.3 Vendor Analysis Team Structure

To ensure accountability and role separation during BCIC execution, the vendor analysis team organisation is shown below.

Figure 1.1: Vendor Analysis Team Organisation Structure



2. Requirements Analysis Using BCIC

Studex Pte. Ltd. conducted a structured vendor-side analysis of the Project Specification using the **Breakdown, Clarify, Interpret, and Categorise** (BCIC) method. The purpose of this analysis was to evaluate requirement quality, scope stability, and readiness for transition into the Software Requirements Specification (SRS).
The analysis focused on identifying ambiguity, mixed responsibility, missing operational bounds, and risks that could affect verification or acceptance. Requirements assessed as internally cogent were preserved unchanged and are reflected in our repository [Link to Github].

2.1 Breakdown of Identified Requirement Issues

Our Business Analyst conducted a thorough analysis of the Project Specification (PS) by applying the Internal Requirements Checklist (IRC) to systematically identify and address any ambiguities or issues within the requirements. The evaluation was performed across four key dimensions: **IRC Dimension**(criteria for evaluating whether a requirement is well-structured, clear, and feasible)**, Evaluation Criteria** (define the specific standards or measures used to assess each IRC)**, Indicator of Sound Requirement** (identifies the signs or markers that a requirement is considered sound or well-formed.) and **Indicator of Incongruence**(problems or deficiencies present in a requirement that may lead to ambiguity, inconsistency, or unfeasibility)**.** The table below outlines the IRC checklist used to assess the clarity, consistency, and feasibility of the requirements.

2.1.1 Incongruous Requirement Checklist

IRC Dimension	**Evaluation Criteria**	**Indicator of Sound Requirement**	**Indicator of incongruence**
Imperative Strength	Appropriateness of modal verb usage	“Shall” is used for mandatory behavior	Weak or inconsistent modality (**“will”, “should”**) used for mandatory logic
Clause Atomicity	Single behavioural responsibility per clause	One atomic behavioural statement	Multiple behaviours combined within one clause
Verifiability	Objective testability of requirement	Clear, measurable acceptance condition defined	No objective test condition can be derived
Responsibility Clarity	Explicit ownership of action	Actor/system authority clearly assigned	Ambiguous, shared, or undefined responsibility
Operational Boundedness	Presence of measurable limits and constraints	Quantified limits (frequency, thresholds, conditions) stated	Missing bounds, undefined limits, or uncontrolled behaviour

Cross-Requirement Consistency	Alignment across BOR and TR clauses	No contradiction or redefinition of prior behaviour	Conflicts, redefinition, or cross-layer misalignment

2.1.2 Identified Issue

A total of 7 requirements are found to be incongruent. As such, these incongruent requirements are also documented in our [github issues]() tab where our team can track and monitor the changes and clarification state of the incongruent requirements.

Affected Requirement	Reason for Incongruence	IRC Dimension
BOR2-L1.1 (Event Notification Management)	Combines notification trigger timing, delivery channels, and user preference behaviour within a single clause, obscuring operational bounds.	Clause Atomicity
BOR2-L1.2 (Hours Completion Alerts)	Specifies notification purpose but lacks enforceable limits on frequency and suppression conditions.	Operational Boundedness
BOR3-L1.1 (Track Event Participation)	Does not clearly distinguish between system-automated tracking and administrative validation of attendance.	Responsibility Clarity
BOR3-L1.2 (Display Event History)	Event history display is specified without explicit linkage to validated attendance states.	Operational Boundedness
TR1-L1.1 (System Classification), TR1-L1.3 (Assess Model), TR1-L1.4 (Architectural Model)	Technical constraints risk redefining BOR-defined operational behaviour.	Cross-Requirement Consistency

2.2 Clarification via Controlled Client Engagement

The **Clarify** step addressed those requirements where intent, authority, or operational bounds could not be conclusively resolved through document analysis alone. Clarification was required to prevent assumption-driven interpretation and to maintain scope stability during SRS derivation.

2.2.1 Account Manager

Our Account Manager, Madeleine, is a Senior Account Manager at Studex Pte. Ltd. with over 12 years of experience managing enterprise software engagements across education and service-sector organisations. She specialises in stakeholder governance, scope control, and

contractual risk management during requirements analysis and delivery transitions. Her role focuses on aligning vendor obligations with institutional expectations while safeguarding delivery feasibility and long-term client relationships.

She is the primary point of contact between the vendor and the customer. As such, to clarify the concerns stated above regarding the incongruent requirements, she carried out the following activities.

1. **Facilitate Clarification Session with Customer**: The Account Manager set up a focus group session with the customer and relevant stakeholders. She arranged the necessary logistics and ensured that the right people were involved in the discussion.
   A pre-agreed agenda was passed to the stakeholders to ensure the discussions were focused on addressing the incongruent requirements. Models such as the 2d napkin were used to illustrate the more complex/priority requirements. This helped to promote discussion and avoid assumptions,s and gain a better understanding.

2. **Escalation Point for issues:** During the focus group session, a few requirements-related conflicts arose. Our Account manager helped to mediate between the customer and the vendor team and helped to resolve misunderstandings.

The session with the client and relevant stakeholders provided valuable insights into their requirements. As a result, we were able to resolve most of the incongruent requirements, while leaving some as open cases that require further clarification from the client. These outcomes were mutually agreed upon and documented at the conclusion of the focus group session. The diagram below illustrates the results of the session, highlighting the requirements that were clarified and those that remain open for further clarification.

2.2.2 Clarification Outcome

Incongruent Requirements	Status (Open/Resolved)	Clarification Summary
BOR2-L1.1 (Event Notification Management)	Resolved	Too many requirements nested into 1 requirement. Discussion with the customer revealed that they want the requirements to be separated and broken down into simpler and easy to track requirements.
BOR2-L1.2 (Hours Completion Alerts)	Open	Clarified with the customer that they want a specific number of notification frequency limits per action. Further details on the numbers and notification method will be specified in the SRS.

BOR3-L1.1 (Track Event Participation)	Resolved	The customer clarified that attendance recording and attendance validation represent two separate authorities. Automated logging reflects raw participation data, but only validated records may represent official completion status. This separation preserves administrative oversight and prevents system assumptions from overriding human verification.
BOR3-L1.2 (Display Event History)	Resolved	The customer confirmed that event history must reflect only validated attendance states to maintain academic integrity. Displaying unverified participation would create reputational and compliance risks. Therefore, history views are restricted to administratively confirmed records.
TR1-L1.1 (System Classification), **TR1-L1.3** (Assess Model), **TR1-L1.4** (Architectural Model)	Resolved	**TR1-L1.1:** The customer required that system classification and architectural decisions must support not redefined business-defined workflows. Role-based access control must align with BOR governance boundaries to ensure administrative authority remains enforceable. **TR1-L1.3:** The customer clarified that technical assessment models must operate within business-defined constraints. Architectural optimisation cannot modify attendance validation logic or notification governance rules without formal approval **TR1-L1.4:** The customer required that the architectural model must enable enforcement of BOR-defined validation and notification controls. Technical layering must support business authority separation rather than collapse it.

2.2.3 Clarification Models Used

To support clarification discussions and prevent assumption-driven interpretation, informal visual models were employed during stakeholder engagement sessions.
The following artefacts were used:

- 2D “napkin-style” workflow sketches illustrating:
  - Attendance validation authority separation
  - Notification trigger boundaries
  - Participation hour allocation logic

- Simple interaction flow prototypes to demonstrate:
  - Dashboard notification behaviour
  - Event validation and record visibility conditions

These models were used solely for clarification and alignment purposes. They were not implementation artefacts and did not imply architectural or UI commitment.

The clarified behaviours reflected in these models were subsequently formalised during the Interpret phase.

Figure 1.2: Clarification Model – Attendance Validation Flow (2D Napkin Illustration)

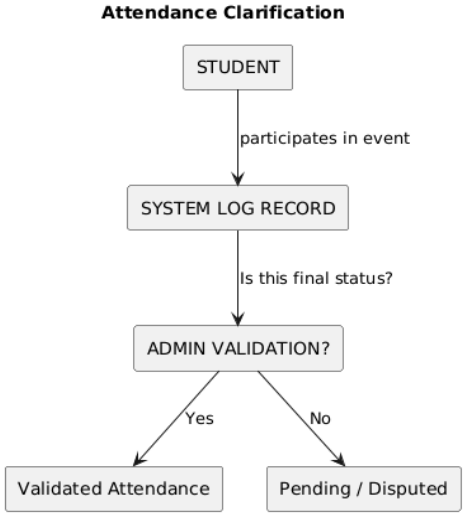
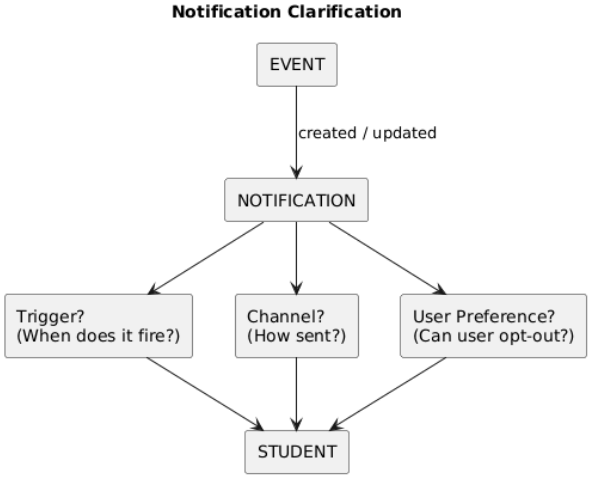


Figure 1.3: Clarification Model – Notification Behaviour Flow (2D Napkin Illustration)



2.3 Interpretation for Specification Readiness

The Interpret phase analyzes the clarified requirements from the Clarify phase, refining them into clear, actionable, and testable specifications. It resolves any remaining ambiguities and ensures the requirements align with the project's objectives. Some incongruent requirements were left open for further clarification, which will be addressed in subsequent discussions. This phase ensures the requirements meet the client's expectations and are ready for the specifying phase.

Original Requirement ID	Revised Cogent Requirement(s)

BOR2-L1.1 (Event Notification Management)	**BOR2-L1.1a:** The system shall trigger a notification when an event is created or updated. **BOR2-L1.1b:** The system shall deliver notifications via defined delivery channels. **BOR2-L1.1c:** Users shall be able to configure notification delivery preferences where permitted.
BOR2-L1.2 (Hours Completion Alerts)	Case is still open
BOR3-L1.1 (Track Event Participation)	**BOR3-L1.1a:** The system shall record student attendance based on participation logs. **BOR3-L1.1b:** Administrative staff shall validate attendance records prior to finalisation. **BOR3-L1.1c:** Validated attendance records shall serve as the authoritative source of participation status.
BOR3-L1.2 (Display Event History)	**BOR3-L1.2a:** Event history displays shall reflect only validated attendance records. **BOR3-L1.2b:** Unvalidated or disputed attendance shall not be reflected as confirmed participation.
TR1-L1.1 (System Classification), **TR1-L1.3** (Assess Model), **TR1-L1.4** (Architectural Model)	**TR1-L1.1:** The system architecture shall support role-based access control aligned with BOR workflows. **TR1-L1.3:** Technical constraints shall not redefine business-defined attendance or notification workflows. **TR1-L1.4:** System classification shall enable enforcement of BOR-defined validation rules.

3. Repository Overview

A Repository has been created to serve as a reference to where all the revised requirements and supporting documents are stored, and it plays an essential role in maintaining traceability, version control, and transparency. It is organised into dedicated folders for Requirements, Models and Diagrams, BCIC, References, Project Documentation and a README folder.

3.1 Repository Organization

- **/Requirements** - stores the revised elicited requirements organized into subfolders for Business operations, Technical requirements and constraints.
- **/Models and Diagrams -** Store all the models related to the project, including both customer-provided models and vendor-created models.
- **/BCIC** - Stores the analysis of the requirements using BCIC and the IRC.
- **/References** - Contains the original Project Specification and any supporting documents
- **/ProjectDocumentation -** Stores the full documentation of the analysis.(includes meeting notes, feedback from clients)
- **/README -** contains details on how to navigate the repository

Version Control - GitHub for its robust version control, easy collaboration, and security features, making it ideal for managing and tracking project requirements and documentation.
Sanity Check - Repo is opened to all members to sanity check the whole set of requirements before the specifying phase.

4. Risk Analysis

Studex Pte. Ltd. identified risk exposure during requirements analysis using the course method: the vendor's eight risks were ranked by the likelihood of happening and the severity of damage to requirements capture.
The Table below records the Risk Analysis for the vendor.
Likelihood of Happening: 1 = Highest, 8 = Lowest
Severity of Damage to Requirements Capture: 1 = Highest, 8 = Lowest

Risk ID	Risk	Likelihood of happening	Severity of damage to requirements capture
R1	Customer changes requirements too much and expects it to be FoC	1	1
R2	Changes are not cross-checked by customer for obvious inconsistencies	2	2
R3	Customer not techie enough to understand explanations about innovative nature of new software	4	6
R4	Customer fights vendor when vendor wants to defer changes that are too big	5	5
R5	Customer refuses to provide technical specs of platform/server environment (claiming security concerns)	6	4
R6	Tender bidding team under-estimated time needed for requirements capture	3	3
R7	Account Manager replaced the middle of RE	7	7
R8	Customer's elicitors leave and take insights with them	8	8

We prioritised risks using a balanced judgement of likelihood and severity of damage to requirements capture, where lower rank values indicate higher exposure. R1, R2, and R6 were selected because they score consistently high across both factors and threaten SRS readiness the most.
Based on our analysis, the top 3 risks by likelihood ranking and severity ranking are listed below, along with their mitigation strategies.

Risk ID: R1
Risk: Customer changes requirements too much and expects it to be FoC
How it could affect the project:
Constant changes to core behaviours (e.g., notification timing/preferences or attendance/hour rules) prevent stabilizing a requirements baseline, causing repeated rework of analysed statements. This delays SRS readiness and can lead to delaying of delivery, scope renegotiation, and increased project effort/cost.

Risk Mitigation Strategies:
- **Baseline + change control:** Agree on when requirements are "confirmed" for SRS; changes after that go through control.
- **Clarification vs change:** Clarification to be done and scope change triggers impact discussion (time/scope) before acceptance.
- **Change record:** Every accepted change must record affected requirement IDs and impacted sections/models.

Risk ID: R2
Risk: Changes are not cross-checked by the customer for obvious inconsistencies
How it could affect the project:
Uncross-checked changes can introduce contradictions between requirements, damaging consistency, traceability, and acceptance criteria. This forces reconciliation and re-validation work, which can delay the SRS and downstream delivery.

Risk Mitigation Strategies:
- **Maintaining consistency:** Change should not be accepted until conflicts with existing requirements are resolved.
- **Traceability check:** Each change must reference related requirements/use cases to validate consistency quickly.
- **Acceptance update:** When a requirement changes, update its acceptance/test condition at the same time.

Risk ID: R6
Risk: Tender bidding team under-estimated the time needed for requirements capture
How it could affect the project:
Underestimated RE time creates pressure to take shortcuts, which weakens requirement clarity/testability and increases downstream rework in SRS, increasing effort and triggering feasibility/time trade-offs.

Risk Mitigation Strategies:
- **Scope prioritisation:** Prioritise high-risk/high-impact requirements first (core flows, boundaries, responsibilities).
- **Schedule control:** Escalate schedule risk early; request additional clarification sessions or scope trimming.

- **Quality control:** Maintain an Open Issues list to avoid assumption-based requirements under time pressure.

5. Sign-off

This analysis confirms that the elicited requirements have been sufficiently reviewed from the vendor perspective based on the information available at the time of analysis. Scope, constraints, and assumptions are understood as currently stated, and the requirements are considered stable for progression to subsequent project phases, subject to customer validation of the controlled clarifications documented in this report.

Studex Pte. Ltd. hereby confirms that the analysis phase has been completed and that this Analysis Report is ready for submission to Vendor C staff for review and approval.

Signed:
Name: Tan Wan Xin, Valerie
Role: Lead of Analysis Team
Organisation: Studex Pte. Ltd.
Date: 7 February 202